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Souvik Sarkar

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EDUCATION

- Master of Technology in Computer Science and Engineering** **2026-2028**
Techno Main Saltlake (TMSL)
Obtained CGPA: Expected 2028
- Bachelor of Technology in Computer Science and Engineering** **2022-2026**
Techno India University (TIU)
Obtained CGPA: 8.19 out of 10

PROJECT

- Bi-Directional Sign Language Translator** [[View Project](#)] **September 2026**
- Engineered an end-to-end bi-directional translation framework converting sign language into English text and vice-versa.
 - Extracted high fidelity 3D spatial temporal hand, face and body pose landmarks to optimize feature representation and reduce computational overhead.
 - Tech Stack:** PyTorch, MediaPipe, Hugging Face.
- SAIFGO Clustering Algorithm** [[View Project](#)] **January 2026**
- Engineered a self-adaptive FGO meta-heuristic clustering algorithm to dynamically model high-variance and diverse data distributions.
 - Achieved a **30.78% improvement** in clustering performance and solution quality over standard baseline models.
 - Validated architectural scalability and mathematical robustness by deploying the model on production-grade datasets, including the Indian Aviation Network.
 - Tech Stack:** Python, Scikit-learn, SciPy, GeoPandas, Matplotlib.
- FGOIKMeans Hybrid Optimization** [[View Project](#)] **August 2025**
- Designed and implemented a hybrid ML algorithm pairing Fungal Growth Optimization with K-Means to resolve local optima stagnation.
 - Optimized model stability and accelerated convergence rates by eliminating K-Means' sensitivity to random centroid initialization.
 - Secured an **18.92% increase** in clustering quality across standard multi-dimensional benchmark datasets.
 - Tech Stack:** Python, NumPy, Pandas, Scikit-learn, Matplotlib.

SKILLS

Programming	Python, C, C++
Web Technologies	HTML, CSS, JavaScript
Game Development	Pygame
ML & Data Tools	scikit-learn, SciPy, scikit-fuzzy, pandas, NumPy, matplotlib, GeoPandas
Documentation	Google Workspace, L ^A T _E X, Microsoft Office
Technical Tools	GitHub, VS Code